

ADITYA KOTHARI

Stanford, CA | +1 (408) 688-8878 | kothari1@stanford.edu | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

Stanford University — Multi-Robot Systems Lab

Sep 2025 – Mar 2027

MS in Aeronautics and Astronautics — GPA: 4.0/4.0

Stanford, CA

- NSF Scholar 25 | KC Mahindra Scholar 25 | JITO JEAP Scholar 25

Vellore Institute of Technology

Jul 2021 – Jul 2025

BS in Mechanical Engineering — GPA: 4.0/4.0

Vellore, India

- Merit Scholarship Awards 2022–2025 | Student Achiever Awards 2024 & 2025

TECHNICAL SKILLS

- **Robotics & Autonomy:** Visuomotor Policies, Multi-Agent Swarm Planning, Agentic Orchestrators, Multi-Object Tracking, SLAM, Path Planning (A^* , θ^* , DWA).
- **Perception & Learning:** Sim2Real, VLMs, Multi-Stage Vision and AI Pipelines, PyTorch, Reinforcement Learning, EKF/IMM
- **Domain Knowledge:** Optimal Control, Sensor Fusion (LiDAR/RGB-D/Vision), State Estimation, GPS Denied Navigation.
- **Tools & Infrastructure:** ROS2, Docker, 3D Gaussian Splatting, Git, Python, C++

ENGINEERING EXPERIENCE

Archer Aviation

Jun 2026 – Aug 2026

Autonomy Intern

San Jose, CA

- Architected an aerial imagery-based multi-agent planning pipeline for **city-scale probabilistic search**, successfully deploying it into Archer's core production autonomy stack.
- Engineered a novel Spectral Multi-Scale Coverage (SMC) planner that uniquely guarantees swarm-level deconfliction at extreme speeds, achieving **>96% joint search optimality** for **100-drone swarms** with **<5s** cycle times.
- Executed the first real-world flight tests and validation of the Archer Autonomy stack on hexacopter platforms, while concurrently committing critical fixes and improvements to the broader autonomy architecture.

Stanford Multi-Robot Systems Lab

Jan 2026 – Present

Student Researcher

Stanford, CA

- Spearheading the development of the AORA **agentic orchestrator**, establishing a closed-loop self-improvement pipeline for robot autonomy via **autonomous policy training** through a novel **agentic Forge** with full **sim2real** deployed in the cloud.
- Designed a system-2 architecture **visuomotor policy** for complex **language-guided drone navigation** tasks with **sim2real** eliminating the need for 60+ hours of manual flying.

Team Veloce (University UAV and Robotics Team)

Apr 2022 – May 2024

Team Captain & Technical Lead

Vellore, India

- Directed a 25-engineer team through the full hardware-software lifecycle to develop and deploy 7 custom UAVs and ground robots for field testing.

ROBOTICS & AUTONOMY PROJECTS

V-LEAD: Vision-Only Drone Navigation in Gaussian Splats | PyTorch, 3D-GS, TD3+BC/SAC, Dagger

Jun 2026

- Distilled a privileged MPC into an **RGB-only visuomotor policy** (ResNet-18 + frozen **Depth Anything V2** cross-attention, GRU, 10-step velocity horizon @ 20 Hz) for goal-conditioned flight in **3D Gaussian Splat** scenes; **BC** → **MPC-oracle Dagger** → **RL** lifted goal success **14–52% → 75–97%** and cut collisions **33% → 5%**.
- Owned the 3DGS sim integration, expert-data pipeline, and **offline-to-online RL** stack; residual **TD3+BC** was the **only** method to beat the imitation seed on an **unseen goal object** (**65.1%** vs 62.4%), where PPO regressed to 40%.

MAVBE: Behavior-Aware Multi-Object Tracking | IMM-EKF, DeepSORT, YOLOv9, CARLA, Python

Mar 2026

- Implemented a behavioral **Interacting Multiple Model (IMM)** filter blending CV/CT/CA dynamics with **social force augmentation** in 3D world-frame coordinates within a modified **DeepSORT** pipeline.
- Achieved **55% RMSE reduction** (0.50 m vs. 1.10 m) and **68% fewer identity switches** over a vanilla Kalman filter baseline at 5-pedestrian density in CARLA; validated on real-world video.

StarNav: GPS-Free Lunar Rover Localization | Particle Filter, OpenCV, Astropy, Python

Mar 2026

- Built a celestial navigation pipeline using multi-threshold **star detection**, geometric triangle signature extraction, and a **particle filter** for position estimation across a 200 km lunar search region with sky imagery alone.
- Achieved $\approx 1^\circ$ **localization accuracy**; improved star detection and expanded triangle signatures, reducing the search space by **99.7%** (38M km² to 126k km²).

Autonomous Ground Robot: On-Robot Autonomy Stack | ROS2, C++, Gazebo

Sep 2025

- Architected a modular **ROS2/C++** stack with **EKF** sensor fusion, Particle Filter (MCL) global localization, and **YOLO**-based perception informing a hybrid **A*/DWA** planner for cluttered dynamic environments.

ACHIEVEMENTS

Rank 1: GKN Aerospace Sustainable Aviation Challenge - Autonomous eVTOL (INR 300,000 Prize)

Nov 2023

Rank 3: AIRBUS National Flight Challenge - Vision Based Brake Safety System

Feb 2024